Last time we defined by definite integral $\int_a^b f(x)dx$ of an integrable function f on $[a,b]$ by

$$\int_a^b f(x)dx = \underline{\int_a^b} f(x)dx = \overline{\int_a^b} f(x)dx,$$

where

$$\underline{\int_a^b} f(x)dx := \sup\{L(f,P) : P \text{ is a partition of } [a,b]\},$$
$$\overline{\int_a^b} f(x)dx := \inf\{L(f,P) : P \text{ is a partition of } [a,b]\}.$$

Here we check that this definition satisfies the following four axioms:

(I0) A continuous function is integrable.
(I1) If $f = c$ is a constant function, then $\int_a^b c = c(b-a)$.
(I2) If $f_1(x) \le f_2(x)$ for all $x \in [a,b]$, then $\int_a^b f_1 \le \int_a^b f_2$.
(I3) If $a \le c \le b$, then $\int_a^b f = \int_a^c f + \int_c^b f$.

1. Verify that $\int_a^b f(x)dx$ satisfies Axiom I1. Start by letting P be any partition of $[a,b]$. Calculate m_i and M_i for each subinterval $[x_{i-1},x_i]$, and work out $\underline{\int_a^b} f(x)dx$ and $\overline{\int_a^b} f(x)dx$ from their definitions to check that they are equal.

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2. Verify that $\int_a^b f(x)dx$ satisfies Axiom I2. Start by assuming $f_1, f_2 : [a,b] \to \mathbb{R}$ are any two integrable function with $f_1(x) \leq f_2(x)$ for all $x \in [a,b]$. For each $i = 0, \ldots, n$, let

$$m_i^{(1)} := \inf\{f_1(x) : x \in [x_{i-1}, x_i]\},$$
$$m_i^{(2)} := \inf\{f_2(x) : x \in [x_{i-1}, x_i]\}.$$

(a) Show that $m_i^{(1)} \leq m_i^{(2)}$.

(b) Show that it follows from part (a) that $L(f_1, P) \leq \int_a^b f_2(x)dx$.

(c) Since P was an arbitrary partition of $[a,b]$, conclude that $\int_a^b f_1(x)dx \leq \int_a^b f_2(x)dx$.

To verify that $\int_a^b f(x)dx$ satisfies the remaining two axioms, we will use the following equivalent definition of integrability.

Lemma. (Darboux's Integrability Criterion) Let $f : [a,b] \to \mathbb{R}$ be a bounded function. Then f is integrable if and only if for every $\varepsilon > 0$ there exists a corresponding partition P_ε of $[a,b]$ such that

$$U(f,P_\varepsilon) - L(f,P_\varepsilon) < \varepsilon.$$

3. Prove the previous proposition.

   (a) ($\Longleftarrow$) *Hint:* Use the fact that $L(f,P) \leq \underline{\int}_a^b f(x)dx \leq \overline{\int}_a^b f(x)dx \leq U(f,P)$ for *every* partition P of $[a,b]$.

   (b) ($\Longrightarrow$) *Hint:* Given $\varepsilon > 0$, begin by choosing partitions P and P' such that $U(f,P) < \overline{\int}_a^b f(x)dx + \frac{\varepsilon}{2}$ and $\underline{\int}_a^b f(x)dx - \frac{\varepsilon}{2} < L(f,P')$.

      i. Explain why P and P' exist.

      ii. Show that $U(f,P) - L(f,P') < \varepsilon$.

      iii. Define a partition P_ε in terms of P and P' so that $U(f,P_\varepsilon) - L(f,P_\varepsilon) < \varepsilon$. (*Hint:* Use the Refinement Lemma.)

4. Verify that $\int_a^b f(x)dx$ satisfies Axiom I3.

   (a) First show that f is integrable on $[a,b]$ if and only if the restrictions of f to $[a,c]$ and $[c,b]$ are integrable. ($\Longrightarrow$) To get started, note that if $c = a$ or $c = b$ there is nothing to prove, so assume $a < c < b$. Assume f is integrable on $[a,b]$ and let $\varepsilon > 0$. Let P_ε be the partition provided by Darboux's Integrability Criterion, and consider the partition $P_c = P_\varepsilon \cup \{c\}$. Show that $U(f,P_c) - L(f,P_c) < \varepsilon$.

   (b) Let $P_1 = P_c \cap [a,c]$ and $P_2 = P_c \cap [c,b]$. Then $P_c = P_1 \cup P_2$. Show that

$$L(f,P_c) = L(f,P_1) + L(f,P_2),$$
$$U(f,P_c) = U(f,P_1) + U(f,P_2).$$

   (*Hint:* Let k be the smallest index such that $x_{k-1} \le c \le x_k$, then break up the sums $L(f,P_c) = \sum_{i=1}^n m_i(x_i - x_{i-1})$ and $U(f,P_c) = \sum_{i=1}^n M_i(x_i - x_{i-1})$.)

   (c) Show that $U(f,P_i) - L(f,P_i) < \varepsilon$ for $i = 1,2$, which shows that the restrictions of f to $[a,c]$ and $[c,b]$ are integrable.

(d) ($\Longleftarrow$) Given $\varepsilon > 0$, use Darboux's Integrability Criterion to produce partitions P_1 of $[a,c]$ and P_2 of $[c,b]$ such that $U(f,P_i) - L(f,P_i) < \frac{\varepsilon}{2}$ for $i = 1,2$. Take $P = P_1 \cup P_2$ and show that $U(f,P) - L(f,P) < \varepsilon$.

(e) We have now proved that f is integrable on $[a,b]$ if and only if the restrictions of f to $[a,c]$ and $[c,b]$ are integrable. As for the value of the integral, fix $\varepsilon > 0$, let P be any partition of $[a,b]$, and define P_c, P_1, and P_2 as before. Show that

$$L(f,P) \leq \int_a^c f(x)dx + \int_c^b f(x)dx \leq U(f,P)$$

and conclude that $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx$.

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5. We will now use the Bisection Method to prove that $\int_a^b f(x)dx$ satisfies Axiom I0.

   To get started, suppose, by way of contradiction, that f is continuous but not integrable on $[a,b]$. It then follows from Darboux's Integrability Criterion that there exists an $\varepsilon > 0$ such that, for every partition P of $[a,b]$, $U(f,P) - L(f,P) \geq \varepsilon$. It then follows that on at least one of the two subintervals $[a, \frac{a+b}{2}]$ and $[\frac{a+b}{2}, b]$, for every P' of that subinterval, $U(f,P') - L(f,P') \geq \frac{\varepsilon}{2}$. For, if this were not the case, then we could find a partition P_L of $[a, \frac{a+b}{2}]$ and P_R of $[\frac{a+b}{2}, b]$ such that $U(f,P_L) - L(f,P_L) < \frac{\varepsilon}{2}$ and $U(f,P_R) - L(f,P_R) < \frac{\varepsilon}{2}$. Then, on the partition $P = P_L \cup P_R$ of $[a,b]$, as we have seen before we would have $U(f,P) = U(f,P_L) + U(f,P_R)$ and $L(f,P) = L(f,P_L) + L(f,P_R)$, and therefore

$$U(f,P) - L(f,P) = U(f,P_L) - L(f,P_L) + U(f,P_R) - L(f,P_R) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

   which contradicts the assumption that $U(f,P) - L(f,P) \geq \varepsilon$ for every partition P of $[a,b]$. Let $I_1 := [a_1, b_1] = [a,b]$, and choose $I_2 := [a_2, b_2]$ to be the subinterval $[a_1, \frac{a_1+b_1}{2}]$ or $[\frac{a_1+b_1}{2}, b_1]$ on which for every partition P of that subinterval we have $U(f,P) - L(f,P) \geq \frac{\varepsilon}{2}$. Continue in this way to obtain a sequence of nested intervals $I_1 \supseteq I_2 \supseteq I_3 \cdots$ with $\inf\{b_n - a_n\} = 0$.

   (a) Fill in the blanks: For every partition P of I_3, $U(f,P) - L(f,P) \geq$ _____.

   For every partition P of I_4, $U(f,P) - L(f,P) \geq$ _____.

   For every partition P of I_n, $U(f,P) - L(f,P) \geq$ _____.

   (b) Let M_n denote the maximum value of f on I_n and m_n the minimum value of f on I_n. Explain why M_n and m_n exist.

   (c) Show that for any partition P of I_n, $U(f,P) \leq M_n \frac{b-a}{2^{n-1}}$ and $L(f,P) \leq m_n \frac{b-a}{2^{n-1}}$.

   (d) Show that for any partition P of I_n, $M_n - m_n \geq \frac{\varepsilon}{2^{n-1}}$.

(e) By the Nested Intervals Theorem, $\bigcap_{n=1}^{\infty} I_n = \{a\}$, where $a = \sup\{a_n\} = \inf\{b_n\}$. Since f is continuous at a, corresponding to $\frac{\varepsilon}{2(b-a)} > 0$ there exists $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x) - f(a)| < \frac{\varepsilon}{2(b-a)}$. Show that there exists $N \in \mathbb{N}$ such that $I_N \subset (a-\delta, a+\delta)$. (*Hint:* Use the Capture Theorem.)

(f) Explain why there exist x_M and x_m in I_N such that $f(x_M) = M_N$ and $f(x_m) = m_N$.

(g) Since $x_M, x_m \in (a-\delta, a+\delta)$,

$$|f(x_M) - f(a)| = |M_N - f(a)| < \frac{\varepsilon}{2(b-a)},$$
$$|f(x_m) - f(a)| = |m_N - f(a)| < \frac{\varepsilon}{2(b-a)}.$$

Show that $M_N - m_N < \frac{\varepsilon}{b-a}$.

(h) Explain why the result in part (g) proves that f is integrable on $[a,b]$.